

Linear Algebra I Recitation Class: Week 3

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Throughout these notes, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ denotes the sets of rational, real, and complex numbers, respectively. V is a vector space over a field F . If you are not yet familiar with fields, you may keep the examples $F = \mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$ in mind.

Let S be a subset of V , we say a vector v is an F -linear combination of S if there exist $s_1, s_2, \dots, s_n \in S$ and $a_1, a_2, \dots, a_n \in F$ such that

$$v = a_1 s_1 + a_2 s_2 + \dots + a_n s_n.$$

The F -span of S is a set collecting all F -linear combinations of S , denoted by $\text{span}(S)$. When the base field is clear in the context, we tend to drop the F from each notation.

We investigate some properties of spans:

Proposition 1. *Let S be a subset of V , $\text{span}(S)$ is the smallest vector subspace of V containing S , i.e. every vector subspace containing S must contain $\text{span}(S)$.*

Proposition 2. *[FIS03, Exercise 1.4.12] A subset W of V is a subspace of V if and only if $\text{span}(W) = W$.*

Proof. ($\implies$) Since W is a vector space containing W , by the minimality of spans, $W \supseteq \text{span}(W)$, and naturally we have $W \subseteq \text{span}(W)$.

($\impliedby$) Every span is a subspace. □

Proposition 3. *[FIS03, Exercise 1.4.13] If S_1 and S_2 are subsets of V such that $S_1 \subseteq S_2$, then $\text{span}(S_1) \subseteq \text{span}(S_2)$. In particular, $\text{span}(S_1) = V$, then $\text{span}(S_1) = V$.*

Proof. This is in HW2. □

Proposition 4. *[FIS03, Exercise 1.4.14] If S_1 and S_2 are arbitrary subsets of V , then $\text{span}(S_1 \cup S_2) = \text{span}(S_1) + \text{span}(S_2)$.*

Proof. This is also in HW2. □

Proposition 5. *[FIS03, Exercise 1.4.15] Let S_1 and S_2 be subsets of V , then $\text{span}(S_1 \cap S_2) \subseteq \text{span}(S_1) \cap \text{span}(S_2)$.*

Proof. Since $S_1 \cap S_2 \subseteq \text{span}(S_1) \cap \text{span}(S_2)$. By the minimality of spans, $\text{span}(S_1 \cap S_2) \subseteq \text{span}(S_1) \cap \text{span}(S_2)$. □

Remark. Let $V = \mathbb{R}^2$, $S_1 = \{(0, 1), (1, 0)\}$, $S_2 = \{(2, 0)\}$, then $\text{span}(S_1 \cap S_2) = \text{span}(\emptyset) = \{0_V\}$ and $\text{span}(S_1) \cap \text{span}(S_2) = V \cap \text{span}(S_2) = \text{span}(S_2) \neq \{0_V\}$. Therefore, the other side of containment in Proposition 5 does **not** hold in general.

A subset S of V is called linear dependent (over F) if there is a nontrivial relation

$$a_1 s_1 + \dots + a_n s_n = 0$$

for some $s_1, \dots, s_n \in S$ (by nontrivial we mean that not all $a_1, \dots, a_n$ are zero). If S is not linearly dependent (over F), then we say S is linearly independent (over F).

We recall some basic facts about linearly dependent and linearly independent sets:

Proposition 6. *S is a linearly independent subset of V if and only if any element $v \in \text{span}(S)$, v can be written as a **unique** linear combination of S (with all coefficients not being zero).*

Proof. ($\Leftarrow$) Since $0 \in \text{span}(S)$ and 0 has a trivial linear combination, by the uniqueness of linear combination (this is our assumption) there is no possibility that we have a nontrivial relation among S , so S is linear independent by our definition.

($\Rightarrow$) Given $v \in \text{span}(S)$, if

$$v = a_1 s_1 + \cdots + a_k s_k = b_1 s_1 + \cdots + b_k s_k$$

for some $s_1, \dots, s_k \in S$ and $a_1, \dots, a_k, b_1, \dots, b_k \in S$ (here we do a small trick: we add extra 0 terms to make the equation look cleaner to deal with). Then we have the relation

$$(a_1 - b_1)s_1 + \cdots + (a_k - b_k)s_k = 0$$

Since S is linear independent, we must have

$$a_1 - b_1 = a_2 - b_2 = \cdots = a_k - b_k = 0$$

Then the proof is done. □

Proposition 7. *Let S be a subset of V :*

1. *If S is linearly dependent, then any superset of S is linearly dependent.*
2. *If S is linearly independent, then any subset of S is linearly independent.*

Proof. 1. If S is linear dependent, then there exists a nontrivial relation among S , which is also a nontrivial relation among any superset of S .

2. If there exists a subset S_0 such that S_0 is linearly dependent, then S should be linearly dependent by 1, which contradicts the assumption. □

Proposition 8. *Let S be a linearly independent set of V , then for any $v \in V \setminus S$, $S \cup \{v\}$ is linearly dependent if and only if $v \in \text{span}(S)$.*

When $V = F^n$, Let $S = \{s_1, \dots, s_k\}$ be a subset of V , where

$$s_i = \begin{pmatrix} a_{1i} \\ a_{2i} \\ \vdots \\ a_{ni} \end{pmatrix} \text{ for each } i = 1, 2, \dots, k.$$

The notions of linear span and linear dependence are closely related to systems of linear equations that we are familiar with from high school algebra.

Linear algebra (abstract)	High school algebra (concrete)
$v = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} \in \text{span}(S)$	$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1k}x_k = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2k}x_k = b_2 \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nk}x_k = b_n \end{cases} \quad \text{has a solution.}$
S is linear independent	$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1k}x_k = 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2k}x_k = 0 \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nk}x_k = 0 \end{cases} \quad \text{has NO nontrivial solution.}$

Spoiler alert: Let $S = \{s_1, \dots, s_n\}$ be a finite set which is linearly independent, from Proposition 6, every element $v \in \text{span}(S)$ can be written uniquely as

$$v = a_1s_1 + \cdots + a_ns_n \quad a_1, \dots, a_n \in F$$

Therefore, we can identify

$$v = a_1s_1 + \cdots + a_ns_n \longleftrightarrow \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \in F^n.$$

In particular, if $V = \text{span}(S)$, then the set S is called an **F -basis** of V , and V can be identified with F^n through the corresponding coordinate map. This is the basic bridge between an abstract vector space and the concrete vector space F^n .

A vector space that has a finite spanning set is called a **finite-dimensional vector space**. A basis of V is a linearly independent spanning set of V , and the number of elements in a basis is called the **dimension** of V .

During the first semester of the linear algebra course, we will explore this bridge and build a dictionary between abstract finite dimensional vector spaces and the more concrete space F^n .

References

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- [Fra02] John B. Fraleigh. *A first course in abstract algebra*. Addison-Wesley Publishing Co., Reading, Mass.-London-Don Mills, Ont., 2002.
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